

EXCEL FORMULAS MASTER CHEAT SHEET

From Beginner to Pro Level — Curated for MIS & Data Operations Interviews

1. Beginner Level: Data Cleaning & Basics

SUM

Definition: Adds all the numeric values in a selected range of cells.

Usage: Used daily to find grand totals of columns like total sales, quantities, or expenses.

Structure: `=SUM( range )`

Example: `=SUM( C2 : C15 )`

COUNTA

Definition: Counts the number of cells in a range that are not empty (counts both text and numbers).

Usage: Used to find out exactly how many rows of data are filled in a sheet.

Structure: `=COUNTA( range )`

Example: `=COUNTA( A2 : A100 )`

TRIM

Definition: Removes all extra leading, trailing, and double spaces from a text string.

Usage: Essential for data cleaning. Raw entry often has accidental spaces that break lookups.

Structure: `=TRIM( text )`

Example: `=TRIM( B2 )` (Converts "Rahul Sharma " to "Rahul Sharma")

PROPER

Definition: Capitalizes the first letter of each word and converts all other letters to lowercase.

Usage: Used to clean up and standardize names, cities, or job titles for clean reporting.

Structure: `=PROPER( text )`

Example: `=PROPER( D2 )` (Converts "bengaluru, KARNATAKA" to "Bengaluru, Karnataka")

CONCAT

Definition: Combines text strings from multiple cells into one single cell.

Usage: Used to merge data fields, like joining a First Name column and a Last Name column together.

Structure: `=CONCAT(text1, text2, ...)`

Example: `=CONCAT(A2, " ", B2)`

2. Intermediate Level: Conditional Logic & Aggregations

IF

Definition: Checks whether a condition is met, and returns one value if TRUE, and another if FALSE.

Usage: Automatically sorting or flagging data based on business rules.

Structure: `=IF(logical_test, value_if_true, value_if_false)`

Example: `=IF(E2>=10000, "Incentive Eligible", "No Bonus")`

SUMIF

Definition: Adds the values in a range that meet one specific, single criterion.

Usage: Calculating target metrics for a single specific entity (e.g., Total sales for "North" region).

Structure: `=SUMIF(range, criteria, [sum_range])`

Example: `=SUMIF(A2:A50, "North", C2:C50)`

SUMIFS

Definition: Adds the values in a range that meet multiple different criteria simultaneously.

Usage: Core engine of MIS reporting. Allows deep drilling (e.g., Sales for "North" AND sold by "John").

Structure: `=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)`

Example: `=SUMIFS(C2:C50, A2:A50, "North", B2:B50, "John")`

COUNTIF

Definition: Counts the number of cells within a range that meet a single specific condition.

Usage: Finding frequencies, such as counting how many tickets are marked as "Pending".

Structure: `=COUNTIF(range, criteria)`

Example: `=COUNTIF(F2:F100, "Pending")`

COUNTIFS

- Definition:** Counts the number of cells within specified ranges that meet multiple conditions.
- Usage:** Finding overlapping metrics, like counting orders from "Bengaluru" AND with "Urgent" status.
- Structure:** `=COUNTIFS(criteria_range1, criteria1, criteria_range2, criteria2, ...)`
- Example:** `=COUNTIFS(D2:D100, "Bengaluru", E2:E100, "Urgent")`

3. Advanced Level: Data Lookup & Structural Matching

VLOOKUP

- Definition:** Searches for a specific value in the first column and returns a value from another column in the same row.
- Usage:** Pulling information (like Email or Salary) from a master table using a unique Identifier / Key.
- Structure:** `=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`
- Example:** `=VLOOKUP(A2, Sheet2!A:D, 3, FALSE)`

INDEX & MATCH (Combined)

- Definition:** MATCH locates the position of a specific item, and INDEX retrieves the value from that position.
- Usage:** The gold standard alternative to VLOOKUP. It can look left, and uses less memory on huge datasets.
- Structure:** `=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`
- Example:** `=INDEX(A2:A100, MATCH(E2, B2:B100, 0))`

IFERROR

- Definition:** Evaluates a formula, and if it results in an error code, outputs a fallback value instead.
- Usage:** Keeping reports clean and presentable by hiding ugly error strings like #N/A or #DIV/0!.
- Structure:** `=IFERROR(value, value_if_error)`
- Example:** `=IFERROR(VLOOKUP(A2, B:D, 2, FALSE), "Record Not Found")`

4. Pro Level: Modern Excel & Dynamic Arrays

XLOOKUP

Definition: Searches a range for a matching item and extracts the corresponding value from a target range.

Usage: The modern VLOOKUP replacement. It looks left or right, defaults to exact match, and doesn't break if rows/columns shift.

Structure: `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])`

Example: `=XLOOKUP(A2, Sheet2!A:A, Sheet2!C:C, "Missing ID")`

UNIQUE

Definition: Extracts and outputs an ordered list of distinct, non-repeating values from a specified range.

Usage: Instantly cleaning out duplicate entries to generate a clean, dynamic summary master list.

Structure: `=UNIQUE(array)`

Example: `=UNIQUE(B2:B500)`

FILTER

Definition: Filters an entire data array based on a logic rule, dynamically outputting every matching record.

Usage: Creating automated, live-updating sub-reports without clicking manual filter dropdowns.

Structure: `=FILTER(array, include, [if_empty])`

Example: `=FILTER(A2:D100, B2:B100="Bengaluru", "No Records")`

CRITICAL INTERVIEW TIP: SUMIF vs SUMIFS Argument Order

Interviewers frequently ask this to spot fake resumes. In **SUMIF**, the range you want to sum is at the **END** of the formula. In **SUMIFS**, the range you want to sum is at the absolute **BEGINNING**. Pro Tip: Always use SUMIFS even for a single condition to keep your spreadsheets future-proof!